CACTUS FALL 2026 · WEEK 2 · LEARNING SESSION

Clean it, chart it, check what the AI wrote

Code can run without an error and still give the wrong answer. Today you learn how to tell.

Kokil Jaidka · National University of Singapore

RECAP OF WEEK 1

What carries over

Your project card

A case study, a main question, and what one row of your data is.

Lists and dictionaries

A list is ordered. A dictionary is labeled slots. A row is a dictionary.

Predict first

Say what you expect before you run. The gap is where you learn.

CONCEPT

A table: rows are things, columns are facts

id	date	comment	country	tone_pos
c01	2024-10-05	Love the new oat latte	SG	6.2
c02	2024-10-05	Queue was too long today	SG	0.0
c07	2024-10-07	Great coffee	TH	7.1

One **row** is one thing: here, one comment. One **column** is one fact about every row. Your project card says what a row is for you.

LIVE DEMO 1 · INSTAGRAM POSTS ABOUT A BRAND

Prompt with no context

```
Write code to get the average views per post.
```

What does the AI have to guess? The file name, the column names, even whether the data is a table at all.

Run it. Read the answer line by line. What would you not be able to check?

LIVE DEMO 2

Prompt with context

```
I have a pandas DataFrame called df from a CSV with these columns: (paste list(df.columns)).
Here are the first 3 rows: (paste df.head(3)). Task: the average views per post. Write the
code, then explain each line in one sentence.
```

Column names so it cannot invent them

Sample rows so it sees the types

An explanation so you can read it

LIVE DEMO 3 · THE CODE WAS RIGHT

A correct answer to the wrong question

2,506

average views per post, as the AI computed it

91%

of posts have 0 views in this data

Many posts have no recorded views at all. The average is not wrong, but it does not describe what we meant.

THE HABIT

Three checks for any result

Count the rows

Before and after each step. What changed, and why?

Check by hand

Pick three rows. Work the answer out yourself.

Use a second method

Get the same number a different way. Do they match?

Write each check in your verification log. A check you did not record is a check you will forget.

LIVE DEMO 4

Watch the row count change

```
print(len(df))
df = df.drop_duplicates()
print(len(df))
df.head()
```

len(df) is the number of rows

Before and after shows what a step removed

.head() shows the first rows

LIVE DEMO 5

Read an error from the bottom

```
Traceback (most recent call last): …
NameError: name 'pd' is not defined
```

1. Read the last line first

2. Say it in your own words

3. Guess the fix, then ask the AI

THE IDEA BEHIND THIS WEEK

Silent changes

A step can drop rows, swap values, or merge groups, and print no error. The only way to know is to check what it did.

A filter loses rows

A merge drops unmatched rows

An average hides group sizes

WHAT TO DO NEXT

Before the coding session

Practice page

Silent Changes, about 20 minutes. It ends with a small coding task for Colab.

Required reading

Textbook Chapter 6, sections 6.1 to 6.4, and Chapter 4, section 4.2 on errors.

Bring

A laptop, the Starbucks file ready to upload, and your verification log.

Stuck for five minutes? Read the last line of the error, re-read the step, ask your partner, ask the AI, then ask for help.

APPENDIX

For self-study and questions

Two short concept slides, on types and on messy data. They are not presented in the session. The same ideas are in textbook Chapter 5.

CONCEPT

Types: the same-looking value can behave differently

NUMBER	TEXT	DATE	TRUE / FALSE
`12`	`"12"`	`2024-10-05`	`True`

A number stored as text looks the same and behaves differently. **"1" + "2" makes "12".** A date stored as text cannot be sorted by time.

CONCEPT

Messy data is normal

Missing values

Empty cells, “N/A”, “unknown”, “-”.

Duplicates

The same row twice, or the same post collected twice.

Wrong types

Dates and numbers stored as text.

Uneven groups

Some countries have thousands of rows and others a dozen.

Cleaning is not a failure of the data. It is part of the analysis, and each step needs a reason.